

Haridev Nambood

(U.S. Citizen) | 559-599-8547 | haridevsathishkumar@gmail.com | linkedin.com/in/haridevsathishkumar

SUMMARY

Junior Electrical Engineering student with hands-on experience developing embedded systems, electronics, and control systems through robotics and university research. Skilled in embedded programming, circuit and PCB design, control algorithms, and hardware testing. Experienced in leading technical teams and delivering measurable improvements in robotics and control projects. Seeking a 2027 internship in embedded systems, electronics, control systems, or automation.

EDUCATION

University of Houston, Cullen College of Engineering | Houston, TX
B.S. in Electrical Engineering

Expected Graduation: May 2028
GPA: 3.9

TECHNICAL SKILLS

Programming: C/C++, Python, MATLAB, LabVIEW, ARM Assembly

Embedded & Controls: Automation, Embedded Systems, Firmware, ESP32, Microcontrollers, Real-Time Systems, PID Control.

Hardware: Circuit Design, PCB Design, Analog Electronics, Digital Logic, Hardware Testing, Debugging, Soldering

Tools: Microsoft Excel, PowerPoint, Word, KiCad, NI Hardware, EtherCAT, Fusion 360, AutoCAD, 3D Printing, OpenCV, Outlook

Awards / Honors: High School Valedictorian, Academic Excellence Scholarship

Engineering: Automation Systems, System Integration, Test/Validation, Technical Documentation, Troubleshooting

WORK EXPERIENCE

Robotics Lab, University of Houston | Houston, TX
Undergraduate Researcher

May 2026 - Present

- Developed LabVIEW real-time control software for magnetically actuated microswimmers—small untethered robots propelled through fluid by rotating magnetic fields for blood-clot removal research—with PID feedback and 3-axis magnetic-field control.
- Developed a variable-speed motion profile that reduced trajectory-tracking error by approximately 50%, from ~10 mm to ~5 mm.
- Integrated dual-camera triangulation using OpenCV and path tracking algorithm to achieve >1 mm 3D position-tracking precision.
- Designed and resin-printed 30+ 4 mm microswimmers optimized for controlled magnetic actuation and fluid-dynamics testing.

Army National Guard | Austin, TX
UH-60M Helicopter Repairer (15T)

Oct 2023 - Present

- Performed maintenance, inspections, and troubleshooting on 50+ UH-60M Black Hawk helicopters to support aircraft safety and mission readiness.
- Repaired and replaced hydraulic pumps, APU accumulators, control rods, and rotor blades on UH-60M Black Hawks supporting the 2026 Texas State Mission, following Army technical manuals and maintenance standards.
- Led a squad during phase inspections, coordinating maintenance tasks and documenting aircraft work through the ACN system.

LEADERSHIP

IEEE, University of Houston | Houston, TX
President, Micromouse Robotics Club

Aug 2025 - Aug 2026

- Led a 30+ member robotics organization, coordinating technical workshops, meetings, and budget costs throughout the year.
- Mentored members into designing and building their own custom Micromouse robots using C/C++, embedded systems, custom circuit design with KiCad, and CAD for 3D modeling while ensuring projects satisfaction for the students.
- Developed and presented tutorials on PID control, choosing compatible parts, pathfinding algorithms, and detailed circuit design, while supporting hardware and software debugging.

RELEVANT PROJECTS

Micromouse Autonomous Robot | Houston, TX

Feb 2025

- Built an ESP32-S3 autonomous maze-solving robot integrating four IR sensors, wheel encoders, IMU feedback, and motor drivers for real-time navigation. Developed maze-mapping and pathfinding algorithms that enabled autonomous maze exploration and fastest route planning to the target.
- Implemented a 500 Hz closed-loop motion-control system using encoder and IMU feedback, achieving ≤ 1 mm stopping accuracy and repeatable turning and path execution.

8-Bit Transistor Adder - Personal Project | Houston, TX

Nov 2025

- Designed and built an 8-bit binary adder entirely from discrete transistors, implementing digital logic without integrated logic ICs.
- Cascaded eight full-adder stages to perform multi-bit binary addition and propagate carry signals through the complete circuit.
- Tested and troubleshot each logic stage to verify correct sum and carry outputs across the 8-bit hardware system.

Self-Balancing Robot | Houston, TX

Jun 2025

- Built an ESP32-based self-balancing robot using real-time sensor feedback and closed-loop motor control.
- Implemented and tuned a PID controller to continuously correct robot tilt and maintain balance during operation.
- Integrated ESP-NOW wireless control and an LCD interface for remote operation and real-time system feedback.